CULTIVATION GUIDE | WATER MANAGEMENT

Improved Alternate Wetting and Drying (AWD)

RRDI's water saving technique for irrigated rice, where it suits, where it does not, and how it interacts with fertiliser timing.

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1. Where it applies

Improved AWD is highly suitable for major and minor irrigated areas of the Dry and Intermediate Zones. It is not suitable for saline paddy fields. It works best when practised across a whole yaya or distribution channel, in collaboration between farmers, agriculture officers and irrigation officers.

RRDI reports that it saves two to three irrigations a season and 10 to 20 percent of total water.

2. The procedure

- Two to three weeks after establishment, irrigate the field to 5 cm and let it drain until the water level falls to 15 cm below the soil surface.
- Irrigate again to 5 cm and let it drain as before.
- Continue until the first flower appears.
- Keep 0 to 5 cm of water for two weeks from the first flower.
- Then stop irrigating and let the water level fall to 15 cm below the soil surface.

3. Measuring the water level

Use a perforated PVC tube about 4 inches in diameter and 20 cm long, cut along its wall. Push it into the soil to 15 cm below the surface, removing the mud inside, and mark its position. Measure the water level inside the tube with a ruler. The number of tubes depends on the landscape of the paddy tract, for example three per acre on flat land.

4. Fertiliser and weeds

Apply fertiliser when the soil is moist, and control weeds at land preparation and before starting AWD. This is the practice the RRDI means when it treats water control as part of fertiliser application rather than separate from it (AV-CG-03).